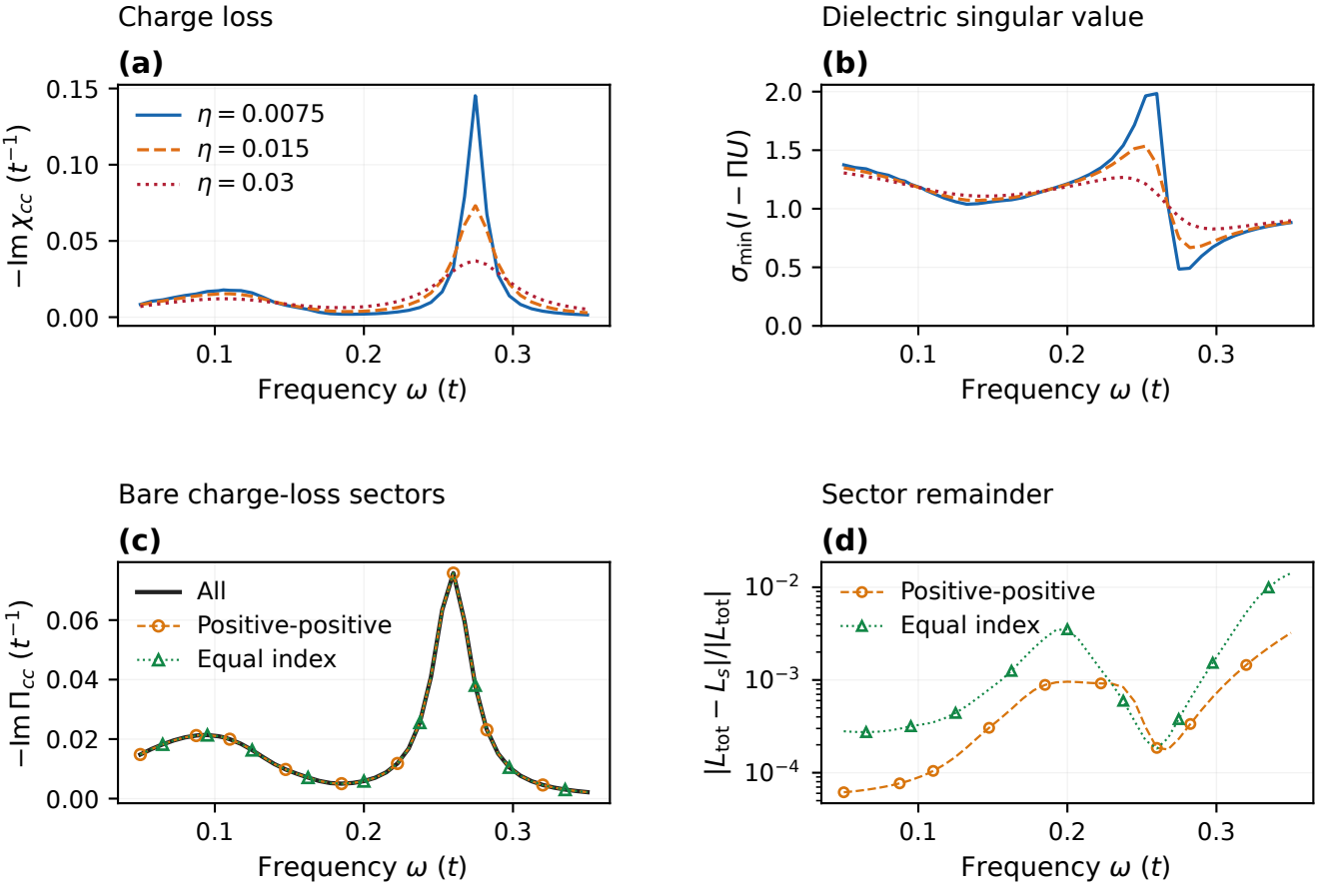Finite-η widths do not certify a gap